

BCA-5001 Knowledge Management

Unit II — Executive Information and Support Systems, Artificial Intelligence, Data Warehousing, OLAP, Data Marts and Business Expert Systems

Semester V | Course Code BCA-5001 | Easy English | Detailed and PYQ-focused

Source and PYQ note: This guide is prepared from the supplied 5001-unit-2.pdf. The year labels below are the year indicators present in that material. Where the source gives a topic/year indicator rather than a verified verbatim paper, it is described as a topic indicator and should be checked against the original university paper.

1. Unit II Roadmap

This unit explains how organizations use information systems, artificial intelligence, data warehouses, OLAP, data marts, and expert systems for analysis and strategic decision-making.

Major area	Core topics	PYQ years indicated
Executive Information and Support Systems	EIS/ESS, DSS construction, DSS role, ESS features and integration	2016, 2020, 2023–24, 2025–26
Artificial Intelligence	Meaning, goals, applications, AI in knowledge management	2023–24, 2024–25, 2025–26
OLTP and OLAP	Differences and uses	2016, 2023–24, 2024–25, 2025–26
Business Expert Systems	Meaning, components, benefits, limitations	2023–24, 2024–25, 2025–26
Multidimensional Analysis	OLAP, data cube and operations	2019, 2020, 2024–25
Data Warehouse	Meaning, need, characteristics, architecture, BI	2019, 2020, 2023–24
Data Mart	Meaning, uses, comparison with warehouse	2024–25, 2025–26
Warehouse design	Top-down and bottom-up approaches	2014, 2015, 2018

2. Executive Information System (EIS) and Executive Support System (ESS)

An Executive Information System (EIS), also called an Executive Support System (ESS), is a specialized information system designed to support senior executives. It provides easy access to important internal and external information required for strategic decisions.

EIS presents information in a simple form such as dashboards, graphs, scorecards, trends, alerts, and summarized reports. It is designed for executives who need a quick understanding of organizational performance rather than detailed transaction records.

Main characteristics

- User-friendly interface with simple menus and dashboards.
- Summarized information focused on KPIs, trends, exceptions, and performance indicators.
- Real-time or near-real-time access to important information.
- Graphical presentation through charts, graphs, heat maps, and scorecards.
- Drill-down capability: an executive can move from a summary figure to detailed data.
- Trend analysis and forecasting using historical and current data.
- Integration of internal data with external information such as competitors, market trends, government policies, and economic indicators.

Main functionalities

- Strategic decision support for investments, mergers, market entry, restructuring, and long-term planning.
- Performance monitoring using sales, profit margin, market share, productivity, and other KPIs.
- Environmental scanning to identify opportunities and threats.
- Personalized dashboards for different executives.
- Data mining and predictive analysis.
- Quick access to critical information during urgent or crisis situations.

Example: A CFO may view revenue, costs, cash flow, and profit margins; a marketing head may view sales, customer acquisition, and market share; the CEO may view an integrated organizational scorecard.

PYQ focus: 2025–26 — What is an EIS/ESS? Explain its features, functionalities, and difference from other information systems.

3. Decision Support System (DSS)

A Decision Support System is a computer-based information system that helps managers make decisions, especially semi-structured and unstructured decisions. It combines data, analytical models, and an interactive user interface.

Steps in constructing a DSS

1. Identification of the problem: Define scope, extent, resources, expected return on investment, risks, and possible conceptual solutions.
2. Decision about the mode of development: Select programming language, development shell, software tools, and technologies suitable for the problem.
3. Development of a prototype: Build an initial working model using available knowledge and interact with domain experts.
4. Prototype validation: Test the prototype, identify weaknesses, and revise it.
5. Planning for a full-scale system: Plan subsystems, assign responsibilities, prepare schedules, and decide implementation resources.
6. Final implementation, maintenance, and evaluation: Implement the complete system, train users, maintain it, and evaluate its performance.

Role of DSS in business

- What-if analysis: Studies the effect of changing input variables.
- Goal-oriented analysis: Determines input values needed to achieve a desired goal.
- Risk analysis: Compares alternatives according to low, medium, or high risk.
- Model building: Represents relationships among variables, assumptions, and constraints.
- Graphical analysis: Uses charts to summarize data, show trends, and compare alternatives.

PYQ focus: 2016 — DSS and its help in business; 2020 — EIS and DSS; 2023–24 — steps in constructing DSS and role in business.

4. EIS/ESS Integration with Other Systems

EIS becomes more useful when it is integrated with other information systems. Integration allows raw operational data, management reports, analytical models, and external information to be transformed into strategic insight.

System	Contribution to EIS
TPS	Provides raw daily transactions such as sales, purchases, payments, and payroll.
MIS	Provides structured routine reports for monitoring and control.
DSS	Provides models, forecasting, simulation, and what-if analysis.

System	Contribution to EIS
ERP	Integrates finance, HR, production, inventory, and other departments.
External information systems	Provide market, competitor, policy, and economic information.

Benefits

- Single and integrated view of the organization.
- Faster decision-making.
- Less manual report preparation.
- Better comparison between actual performance and targets.
- Improved response to external changes.
- More effective strategic planning.

PYQ focus: 2025–26 — Explain how EIS can be integrated with other information systems.

5. Artificial Intelligence (AI)

Artificial Intelligence is a branch of computer science concerned with creating systems that perform tasks requiring human-like intelligence. Such tasks include learning, reasoning, problem solving, perception, language understanding, planning, and decision-making.

Core capabilities of AI

- Knowledge representation.
- Reasoning.
- Problem solving.
- Learning.
- Perception.
- Natural-language understanding.
- Planning.
- Object manipulation and movement.

Goals of AI in knowledge management

1. Automate knowledge capture from documents, emails, databases, and unstructured data.
2. Improve knowledge discovery by finding patterns, relationships, and trends.
3. Support knowledge sharing and collaboration by recommending documents, experts, and communities.
4. Personalize knowledge delivery according to user role, interests, and learning behaviour.
5. Improve search and retrieval using semantic search and knowledge graphs.
6. Automate knowledge maintenance by finding outdated, inconsistent, or incorrect information.
7. Enable predictive analytics for future trends, risks, and proactive actions.

Applications of AI

- Game playing: Systems search possible moves and use heuristics to select actions.
- Speech recognition: Converts spoken language into text or commands.
- Natural-language understanding: Interprets language in a particular domain.
- Computer vision: Extracts meaning from images and video.
- Heuristic classification: Places an object or situation into a category, such as approving a suspicious transaction.

- Expert systems: Provide advice using domain knowledge and rules.
- Chatbots and virtual assistants: Provide real-time access to organizational knowledge.
- Predictive analytics: Forecasts demand, risks, customer behaviour, or market trends.

PYQ focus: 2023–24 — meaning and applications of AI; 2024–25 — AI and its goals/significance in knowledge management; 2025–26 — AI, goals, and applications.

6. OLTP and OLAP

OLTP means Online Transaction Processing. It supports routine operational transactions such as INSERT, UPDATE, and DELETE. OLAP means Online Analytical Processing. It supports complex analysis, aggregation, reporting, and decision-making.

Basis	OLTP	OLAP
Purpose	Daily operations and transaction control	Analysis, planning, and decision support
Data	Detailed and current	Historical, summarized, and integrated
Queries	Short, simple, standardized	Complex, analytical, and aggregation-based
Users	Operational staff and transaction users	Managers, analysts, and executives
Database design	Highly normalized, often 3NF	Often denormalized, star/snowflake schemas
Operations	Frequent inserts, updates, and deletes	Mostly read and analytical operations
Response requirement	Very fast transaction response	Fast analytical response; complex queries may take longer
Source	Original operational data	Usually receives data from OLTP/data warehouse
Backup concern	Operational data loss can stop business	Can often be reloaded from source systems

Example: A college admission system using student registration, fee payment, and attendance updates is OLTP. A report showing admissions by course, year, gender, and district is OLAP.

PYQ focus: 2016, 2023–24, 2024–25, 2025–26 — differentiate OLTP and OLAP.

7. Multidimensional Analysis and OLAP

Multidimensional analysis examines data from multiple perspectives or dimensions, such as time, product, region, customer, department, or sales channel. It is widely used in data warehouses and OLAP systems.

OLAP enables users to extract and view data from different points of view. A relational table is commonly viewed as two-dimensional, but a multidimensional model organizes measures across several dimensions.

OLAP cube

An OLAP cube is a multidimensional data structure. Each cell contains a measure, such as sales, profit, cost, quantity, or budget, at the intersection of dimension members.

Time | | Product ■■■■■■ Data Cube ■■■■■■ Region Example dimensions: Product × Time × City Example measure: Sales amount

OLAP operations

Operation	Meaning	Example
Roll-up	Moves upward in a concept hierarchy; summarizes data	Day → Month → Year

Operation	Meaning	Example
Drill-down	Moves downward; shows greater detail	Year → Quarter → Month
Slice	Selects one value of a dimension	Sales for 2025 only
Dice	Selects a sub-cube using multiple conditions	Sales for two products in two cities
Pivot/rotate	Changes the orientation of dimensions	Rows and columns are exchanged
Drill-through	Moves from aggregate data to source transaction details	Annual sales → individual invoices

Concept hierarchies

A concept hierarchy groups values at different levels. For example, day can be grouped into month and year; city can be grouped into state and country; product can be grouped into category and department.

PYQ focus: 2019 — multidimensional data and data cube operations; 2020 — OLAP; 2024–25 — meaning of multidimensional analysis.

8. Data Warehouse

A data warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data designed to support decision-making. It provides a central and consistent view of current and historical information.

Four classic characteristics

- Subject-oriented: Organized around important subjects such as customers, sales, products, or finance.
- Integrated: Data from different sources is cleaned, standardized, and combined.
- Time-variant: Historical data is stored with a time dimension for trend analysis.
- Non-volatile: Once loaded, data is mainly read and analysed rather than frequently changed like operational data.

Why a data warehouse is needed

1. Centralized storage of data from multiple sources.
2. Improved data quality and consistency through cleansing and transformation.
3. Support for complex analytical queries and decision-making.
4. Historical analysis and trend identification.
5. Scalability for very large data volumes.
6. Integration of data from different formats and systems.
7. Support for business intelligence, dashboards, and reporting.
8. Better query performance without disturbing daily transaction systems.

ETL process

Operational databases / external sources ■ Extract ■ Transform and clean ■ Load ■ Data warehouse ■
OLAP / BI / reports

ETL means Extract, Transform, and Load. Data is extracted from source systems, cleaned and transformed into a common structure, and loaded into the warehouse. Refresh processes update the warehouse periodically or continuously according to the design.

PYQ focus: 2019 — three-tier architecture and characteristics; 2020 — data warehouse architecture; 2023–24 — characteristics and need of a data warehouse.

9. Three-Tier Data Warehouse Architecture

A common data-warehouse architecture contains three tiers.

1. Bottom tier — Data warehouse database server: Stores integrated warehouse data, usually in a relational database. Back-end tools perform extraction, cleaning, transformation, loading, and refresh.
2. Middle tier — OLAP server: Provides multidimensional analysis. It may use ROLAP, MOLAP, or a hybrid approach.
3. Top tier — Front-end client tools: Contains query tools, reporting tools, analysis tools, dashboards, and data-mining tools.

Top tier: Query / Reports / Analysis / Data Mining ■ Middle tier: OLAP Server ■ Bottom tier: Data Warehouse Database ▲ ETL from operational/external sources

Types of data warehouse

Type	Meaning
Enterprise Data Warehouse	Centralized warehouse serving the whole organization.
Operational Data Store	Frequently refreshed store for integrated operational reporting, often nearer to current data.
Data Mart	Smaller, focused subset for a department or subject such as sales or finance.

10. Data Mart

A data mart is a smaller, subject-specific subset of organizational data designed for a particular department, user group, or business function. A sales data mart may contain product, customer, order, and revenue information.

Characteristics and benefits

- Smaller than an enterprise data warehouse.
- Focused on one department or subject.
- Can be developed in weeks rather than months or years.
- Usually less expensive to implement.
- Easier for users to understand.
- Can provide faster reporting because fewer data is queried.
- Can be customized for departmental requirements.
- Can be deployed on smaller or distributed servers.
- Supports incremental development.

Basis	Data warehouse	Data mart
Scope	Enterprise-wide	Department or subject-specific
Size	Very large	Small or medium
Purpose	Strategic integrated view	Focused departmental analysis
Users	Organization-wide	Specific user group
Development	Large and complex	Faster and more focused
Data	Integrated data from many areas	Subset of warehouse or selected source systems

PYQ focus: 2024–25 — data mart and difference from warehouse; 2025–26 — define data mart and compare it with data warehouse.

11. Business Intelligence (BI) in Data Warehousing

Business Intelligence is a collection of strategies, technologies, applications, data, and processes used to analyse business information and convert it into actionable insight.

Importance of BI

- Combines internal data such as finance, sales, customers, and operations with external market data.
- Supports operational decisions such as pricing and product positioning.
- Supports strategic decisions such as goals, priorities, investment, and direction.
- Provides dashboards, reports, alerts, and visual analysis.
- Helps identify trends, patterns, risks, and opportunities.
- Improves the quality and speed of managerial decisions.

Data warehousing and BI work together. The warehouse supplies cleaned, integrated, and historical data, while BI tools provide reports, dashboards, OLAP analysis, and data visualizations. The combined approach is often described as BI and data warehousing.

PYQ focus: 2020 — importance of BI in a data warehouse.

12. Data Transformation

Data transformation is the process of converting, cleaning, restructuring, standardizing, and enriching data so that it becomes suitable for analytics, reporting, storage, or decision-making.

Common transformation activities

- Changing data types and formats.
- Removing duplicate records.
- Handling missing values.
- Standardizing names, units, dates, and codes.
- Combining data from multiple sources.
- Aggregating daily data into monthly or yearly data.
- Filtering irrelevant records.
- Creating calculated fields and derived measures.
- Applying validation rules.

The goal is to make data accurate, consistent, accessible, understandable, and actionable. AI and machine-learning tools can automate parts of data transformation.

PYQ focus: 2025–26 — explain data transformation.

13. Business Expert System (BES)

A Business Expert System is a knowledge-based information system based on artificial intelligence. It stores expert knowledge about a specific problem domain and provides advice or recommendations to users.

An expert system is interactive. It can answer questions, apply rules to facts, and explain the reasoning behind its conclusion. It is useful when a problem is complex and a simple algorithm cannot provide a satisfactory answer.

Components

Component	Function
Knowledge base	Stores domain facts, rules, heuristics, and expert knowledge.
Inference engine	Applies logic and rules to facts to derive conclusions.
User interface	Allows users to enter information and receive advice.
Explanation facility	Shows how or why a conclusion was reached.
Knowledge acquisition module	Helps collect and update knowledge from experts or sources.

Advantages

- Provides consistent answers.
- Preserves important expert knowledge.
- Encourages clarification of decision logic.
- Does not forget routine questions.
- Can work continuously.
- Can serve many users.

Limitations

- Lacks general common sense in unfamiliar situations.
- Cannot always produce creative human-like responses.
- Experts may find it difficult to explain their reasoning.
- Incorrect rules or facts can produce wrong advice.
- Requires updating when the environment changes.

Applications

- Accounting and auditing.
- Law and compliance.
- Medicine and diagnosis.
- Financial services and loan assessment.
- Trading recommendations.
- Production and process control.
- Human-resource advisory systems.

PYQ focus: 2023–24 — definition and components; 2024–25 — business expert system, working, benefits, and limitations; 2025–26 — expert system and components.

14. Top-Down and Bottom-Up Warehouse Design

Two common approaches are used to build a data warehouse: top-down and bottom-up.

Bottom-up approach

In the bottom-up approach, data marts are created first for individual business areas such as sales, finance, or inventory. These data marts are later integrated to form an enterprise data warehouse. Shared dimensions are called conformed dimensions.

Advantages of bottom-up

- Data marts can be delivered quickly.
- Reports can be generated early.
- Development is incremental.
- New departments can be added by creating more data marts.
- Initial cost and scope may be smaller.

Disadvantages of bottom-up

- Different marts may become inconsistent if common standards are not enforced.
- Integration can become difficult later.
- The overall enterprise view may take longer to achieve.

Top-down approach

In the top-down approach, the enterprise data warehouse is built first. Departmental data marts are then created from the integrated warehouse.

Advantages of top-down

- Provides consistent dimensions and definitions.
- Creates an integrated enterprise view.
- Data marts inherit standardized warehouse data.
- Robust for organization-wide strategic reporting.

Disadvantages of top-down

- Large initial project.
- Higher cost and longer implementation time.
- Less flexible for immediate departmental needs.
- Benefits may appear later.

Basis	Bottom-up	Top-down
Starting point	Data marts	Enterprise data warehouse
Delivery	Fast departmental results	Slower initial delivery
Integration	Performed while combining marts	Built into the central warehouse
Cost	Initially lower	Initially higher
Flexibility	Good for incremental growth	Good for enterprise consistency
Main risk	Inconsistent marts	Large and costly project

PYQ focus: 2014 and 2018 — compare approaches and merits/demerits; 2015 — explain two warehouse-building approaches.

15. Complete Year-Wise PYQ Index

The following list records every year/topic indicator found in the supplied 5001 Unit-II material.

Year	Question/topic indicated in supplied material
2014	Top-down and bottom-up approaches for building a data warehouse; merits and demerits.
2015	Two approaches for building a data warehouse with advantages and disadvantages.
2016	Role of DSS/EIS in business; difference between OLTP and OLAP.
2018	Top-down and bottom-up data-warehouse design approaches.
2019	Multidimensional data and data cube operations; three-tier data-warehouse architecture and characteristics.
2020	EIS and how DSS helps business; OLAP; data-warehouse architecture; importance of BI in data warehouse.
2023–24	Steps in constructing DSS and role of DSS; AI applications; OLTP vs OLAP; expert-system components; data warehouse characteristics and need; data cube/OLAP topics.

Year	Question/topic indicated in supplied material
2024–25	AI and goals/significance in knowledge management; data mart and warehouse comparison; OLTP vs OLAP; multidimensional analysis; business expert system.
2025–26	EIS/ESS meaning, features, functionalities, integration, and environmental response; AI and goals; data mart; data transformation; OLTP vs OLAP; expert system and components.

Most repeated/high-priority topics

Priority	Topic	Years indicated
Very high	EIS/ESS, DSS role and construction	2016, 2020, 2023–24, 2025–26
Very high	OLTP vs OLAP	2016, 2023–24, 2024–25, 2025–26
Very high	Data warehouse architecture/need/characteristics	2019, 2020, 2023–24
Very high	AI and knowledge management	2023–24, 2024–25, 2025–26
High	Expert system	2023–24, 2024–25, 2025–26
High	Data mart	2024–25, 2025–26
High	Multidimensional analysis/OLAP/data cube	2019, 2020, 2024–25
High	Top-down vs bottom-up	2014, 2015, 2018

16. Important Questions by Marks

2-mark questions

1. Define EIS.
2. Define DSS.
3. What is AI?
4. What is OLAP?
5. What is OLTP?
6. Define data warehouse.
7. What is a data mart?
8. What is a data cube?
9. Define data transformation.
10. What is an expert system?
11. Expand ROLAP and MOLAP.
12. What is ETL?

5-mark questions

1. Explain the role of DSS in business.
2. Write a note on EIS.
3. Explain the goals of AI in knowledge management.
4. Differentiate OLTP and OLAP.

5. Explain data transformation.
6. Explain data mart and its benefits.
7. Write a note on the components of an expert system.
8. Explain OLAP operations.

10-mark questions

1. Explain EIS/ESS features and functionalities.
2. Explain the construction steps and role of DSS.
3. Explain AI, its goals, and applications.
4. Explain multidimensional analysis and OLAP cube operations.
5. Explain the three-tier data-warehouse architecture.
6. Discuss characteristics and need of a data warehouse.
7. Explain data mart and compare it with a data warehouse.
8. Explain an expert system, components, benefits, and limitations.
9. Compare top-down and bottom-up warehouse design approaches.
10. Explain how EIS integrates with TPS, MIS, DSS, ERP, and external sources.

17. Quick Revision and Exam Strategy

Definitions to memorize

Term	Exam-ready meaning
EIS/ESS	Executive-level system providing summarized internal/external information for strategic decisions.
DSS	Computer-based system helping managers analyse alternatives and solve semi-structured problems.
AI	Computer-science field creating systems with human-like learning, reasoning, perception, and problem-solving abilities.
OLTP	System for fast, routine, detailed operational transactions.
OLAP	System for multidimensional analysis of historical and aggregated data.
Data warehouse	Subject-oriented, integrated, time-variant, non-volatile data collection for decision support.
Data mart	Focused subset of organizational data for a department or subject.
Expert system	AI-based knowledge system that applies facts and rules to provide expert advice.

How to write a 10-mark answer

1. Start with a correct definition.
2. Draw a labelled diagram or architecture where possible.
3. Use meaningful headings.
4. Explain every point in complete sentences.
5. Add examples, advantages, limitations, and comparisons.

6. Mention the relevant PYQ year in your personal revision notes, but do not write the year in an answer unless the question asks for it.
7. End with a short conclusion.

Seven-day plan

Day	Target
1	EIS, ESS, DSS definition, role, and construction steps.
2	EIS features, functionalities, integration, and comparison with MIS/DSS/TPS.
3	AI meaning, goals, applications, and significance in knowledge management.
4	OLTP, OLAP, multidimensional analysis, data cubes, and operations.
5	Data warehouse meaning, characteristics, need, ETL, and three-tier architecture.
6	Data marts, BI, data transformation, and top-down/bottom-up design.
7	Expert systems, complete year-wise PYQ revision, and one mock test.

Final reminder: The year-wise table is based on the supplied 5001-unit-2.pdf. For exact verbatim wording, verify each entry against the original university question paper.